

Anti-Forgetting Transfer Learning for Cross-Power Fault Diagnosis Based on Continual Learning Strategies

Author Name, Affiliation (Conference Draft, Replace According to Submission Requirements)

Abstract—This paper addresses cross-domain performance degradation caused by power-dependent vibration distribution shifts and catastrophic forgetting caused by target-domain fine-tuning. An anti-forgetting transfer learning framework combining a multi-modal CrossViT backbone and continual-learning strategies is proposed. Three-channel vibration signals are constructed as time-domain windows and STFT spectrograms; a dual-branch Transformer encoder and cross-modal attention fusion module are used to extract fault-discriminative features. A teacher model is pre-trained on the 1.0 kW source domain, and knowledge distillation with LWF source replay is introduced during adaptation to the 3.0 kW target domain. Under the latest stride=128 setting, Distill+LWF (T=2, alpha=0.3) achieves 91.03% source retention, 89.10% target accuracy, and a Balance Score of 99.10; Distill (T=1.5, alpha=0.25) further achieves the highest source retention of 94.23% with 87.18% target accuracy. Confusion matrices, training curves, and attention activation maps verify the anti-forgetting capability and interpretability of the proposed framework.

Keywords—cross-power fault diagnosis; transfer learning; catastrophic forgetting; knowledge distillation; continual learning; CrossViT

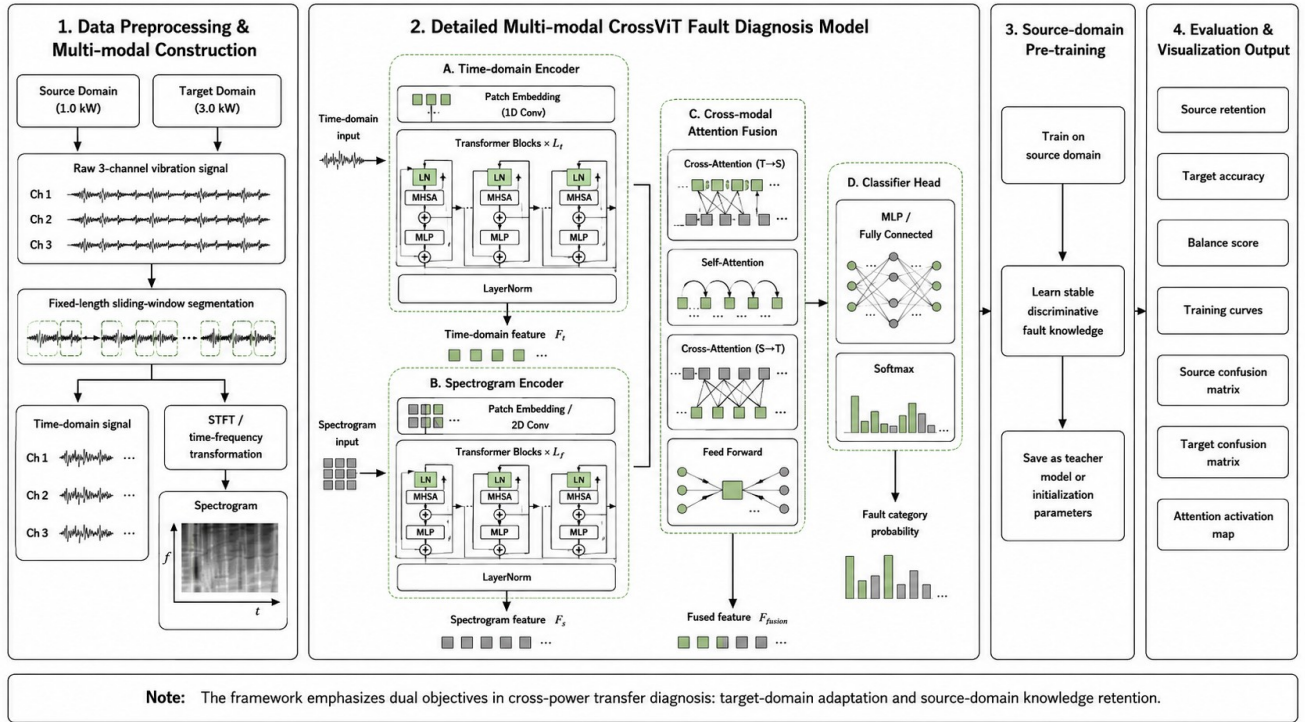


Fig. 1. Overall framework of anti-forgetting cross-power transfer fault diagnosis.

I. Introduction

Fault diagnosis of rotating machinery is a fundamental component of prognostics and health management. Conventional machine-learning methods depend heavily on handcrafted features, while deep neural networks can learn discriminative representations directly from raw vibration signals. Nevertheless, most diagnostic models assume that training and testing data follow the same distribution. When the operating power changes from 1.0 kW to 3.0 kW, the vibration amplitude, energy distribution, and time-frequency structure change substantially, causing a severe degradation of models trained in a single power domain. Deep learning-based PHM has been widely studied because of its representation-learning ability [1], but robust cross-condition deployment remains challenging.

Transfer learning and domain adaptation provide an effective way to reduce cross-domain discrepancy. Existing studies have explored domain-shared CNNs, MMD-based distribution alignment, pseudo-label learning, adversarial adaptation, and Wasserstein distance for transfer diagnosis [2]-[8]. Recent Vision Transformer based dual-domain fusion also demonstrates the complementary value of time-domain and

frequency-domain signals [5]. However, most methods mainly optimize target-domain accuracy and rarely evaluate whether the adapted model still preserves source-domain diagnostic knowledge. In practical equipment, a machine may repeatedly switch among multiple power conditions; thus, a useful model should learn the new domain without forgetting the old one.

This paper investigates catastrophic forgetting in cross-power fault diagnosis and develops an anti-forgetting transfer framework based on a multi-modal CrossViT backbone and continual-learning strategies. The contributions are threefold: (1) a dual-objective evaluation protocol with source retention, target accuracy, and Balance Score is constructed; (2) fine-tuning, freezing, EWC, MMD, DANN, pseudo-labeling, knowledge distillation, and LWF replay are systematically compared; (3) under the latest stride=128 setting, distillation-based methods are validated with confusion matrices, training curves, and attention activation maps.

II. Methodological Background

Transfer learning reuses fault-discriminative knowledge learned from a labeled source domain to improve prediction in a target domain. Let $D_s = \{(x_i^s, y_i^s)\}$ and

$D_t = \{(x_j^t, y_j^t)\}$ denote the source and target domains, where x contains both time-domain signals and spectrograms. The objective is to minimize the target-domain classification loss while avoiding source-domain degradation. MMD/JMMD and DANN are two representative adaptation mechanisms: MMD measures distribution discrepancy in a reproducing kernel Hilbert space, whereas DANN uses a gradient reversal layer to encourage domain-invariant features.

Continual learning focuses on preserving old knowledge when learning new tasks. EWC constrains changes of important parameters using Fisher information, while knowledge distillation transfers teacher knowledge through soft labels. Compared with purely parameter-based regularization, distillation preserves inter-class similarity relations and is therefore attractive for small-sample industrial fault diagnosis with large domain shifts.

III. Proposed Anti-Forgetting Transfer Framework

The diagnostic backbone is a multi-modal CrossViT. Raw three-channel vibration signals are segmented into windows of length 1024 and converted into spectrograms of size $3 \times 128 \times 128$ using STFT. A time-domain encoder with 1D patch embedding and Transformer blocks captures waveform impacts and long-range dependencies. A spectrogram encoder with 2D patch embedding extracts time-frequency energy patterns. A cross-modal attention module performs bidirectional $T \rightarrow S$ and $S \rightarrow T$ interactions, followed by an MLP classifier.

$\min_{\theta} \mathcal{L}_t(\theta)$, subject to preserving source-domain decision knowledge

After source-domain pre-training, the model is saved as a teacher f_T and also used to initialize the student f_S . During target-domain adaptation, the student is optimized with both hard-label cross entropy and teacher-student KL divergence: $L = \alpha \cdot \text{CE}(y, f_S(x)) + (1-\alpha) \cdot T^2 \cdot \text{KL}(\text{softmax}(z_T/T) \parallel \text{softmax}(z_S/T))$.

Here T is the distillation temperature and α is the weight of the hard-label loss. LWF replay further mixes a small ratio of source-domain samples into each fine-tuning epoch. The Balance Score is defined as $B = A_t + 10 \cdot I(A_s \geq \tau)$, where A_t is target accuracy, A_s is source retention, and τ is the source-retention threshold. This score rewards a high target-domain accuracy only when source knowledge is sufficiently preserved.

Algorithm 1 Anti-forgetting cross-power transfer training

1. Input source domain D_s , target domain D_t , CrossViT model f , temperature T , weight α , and replay ratio r .
2. Train f on D_s to obtain the teacher f_T and initial parameters $\theta_{0.0}$.
3. Initialize the student f_S from f_T .
4. For each adaptation epoch, form a target batch B_t and sample a source replay batch B_s with ratio r .
5. Compute cross-entropy loss, teacher-student distillation loss, and total loss L .
6. Update f_S and record source retention, target accuracy, and Balance Score.
7. Output the model with the best Balance Score and generate confusion matrices, curves, and attention maps.

IV. Experimental Setup

The transfer task is 1.0 kW $\rightarrow$ 3.0 kW cross-power fault diagnosis. Each domain contains 394 samples. The temporal input size is (1024,3), the spectrogram input size is (3,128,128), and the number of fault classes is 16. The latest version uses stride=128 for sliding-window segmentation, increasing local pattern coverage. The main model uses embed_dim=384, num_heads=4, num_layers=2, dropout=0.1, batch size 32, and learning rate $1e-4$.

Main experimental results:

- V3-Baseline: source 11.54%, target 93.59%, Balance 93.59; High target but severe forgetting.
- Distill+LWF, $T=2$, $\alpha=0.3$: source 91.03%, target 89.10%, Balance 99.10; Best overall balance.
- Distill, $T=1.5$, $\alpha=0.25$: source 94.23%, target 87.18%, Balance 97.18; Highest source retention.
- Distill, $T=1.8$, $\alpha=0.3$: source 90.38%, target 86.54%, Balance 96.54; Stable distillation setting.
- DANN/MMD methods: source <12.5%, target 72.5%-75.0%, Balance Low; Useful target adaptation but forgetting.

V. Results and Discussion

The latest results show a clear trade-off. Direct fine-tuning achieves a target accuracy of 93.59%, but the source-domain accuracy decreases to only 11.54%, indicating severe catastrophic forgetting. In contrast, Distill+LWF ($T=2$, $\alpha=0.3$) achieves 91.03% source retention and 89.10% target accuracy, yielding the best Balance Score of 99.10. In the V4 distillation search, $T=1.5$ and $\alpha=0.25$ further improves source retention to 94.23% while maintaining 87.18% target accuracy, suggesting that a lower temperature and a moderate hard-label weight help protect source knowledge.

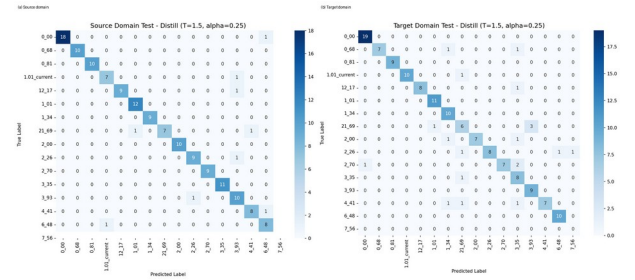


Fig. 2. Confusion matrices of Distill ($T=1.5$, $\alpha=0.25$) on the source and target domains.

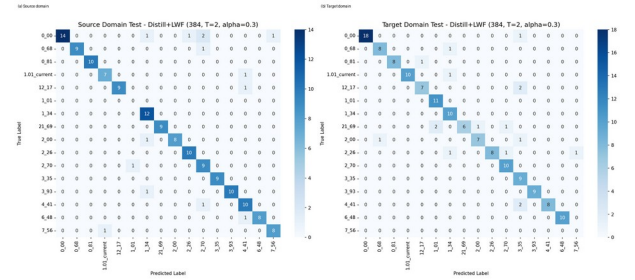


Fig. 3. Confusion matrices of Distill+LWF ($T=2$, $\alpha=0.3$) on the source and target domains.

The confusion matrices exhibit clear diagonal structures in both domains, confirming that the distillation-based models do not simply bias toward the target domain but preserve the separability of most source-domain classes. This is a significant improvement over baseline fine-tuning, whose source-domain confusion matrix is sparse along the diagonal.

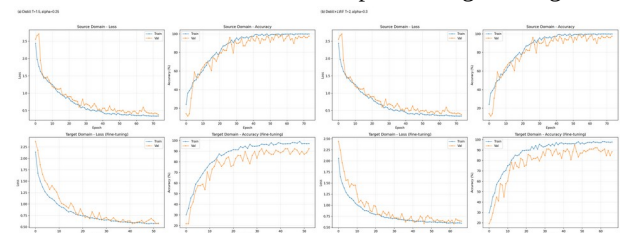


Fig. 4. Source pre-training and target fine-tuning curves of two distillation-based models.

Training curves show that source-domain pre-training converges stably and target-domain fine-tuning improves rapidly without excessive validation instability. Attention activation maps further indicate that the model focuses on impulsive waveform positions, periodic variations, low-frequency spectral bands, and locally enhanced time-frequency regions. The predicted probabilities concentrate near the correct class, improving the interpretability of the diagnostic decision.

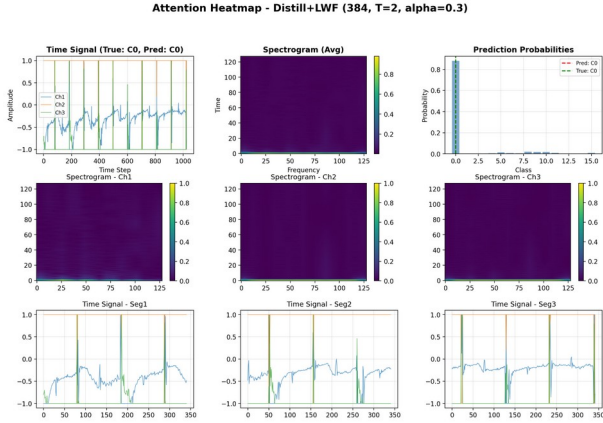


Fig. 5. Attention-based activation visualization: time segments, spectrogram responses, and class probabilities.

VI. Conclusion

This paper presents an anti-forgetting transfer-learning framework for cross-power fault diagnosis. Experiments show that conventional fine-tuning and some domain-adaptation methods improve target-domain accuracy but suffer from severe source-domain forgetting. EWC is less effective in the current small-sample cross-power setting. Knowledge distillation combined with LWF replay achieves the best dual-domain trade-off. Under the latest stride=128 setting, Distill+LWF ($T=2$, $\alpha=0.3$) reaches 91.03% source retention, 89.10% target accuracy, and a Balance Score of 99.10. Future work will extend the framework to multi-power continual transfer, unlabeled target-domain adaptation, and more stable cross-modal attention interpretation.

References

- [1] Y. Zhang and Y.-F. Li, "Prognostics and health management of Lithium-ion battery using deep learning methods: A review," *Renewable and Sustainable Energy Reviews*, 2022.
- [2] H. Cao, W. Luo, H. Shao, F. Zhu, M. Xia, and D. Su, "Unsupervised domain-shared convolutional neural network for bearing fault transfer diagnosis," *ICCSE*, 2021.
- [3] B. Yang, Y. Lei, F. Jia, and S. Xing, "A transfer learning method for intelligent fault diagnosis from laboratory machines to real-case machines," *SDPC*, 2018.
- [4] Z. Du, B. Yang, Y. Lei, X. Li, and N. Li, "A hybrid transfer learning method for intelligent fault diagnosis of machines under variable operation conditions," *PHM-Qingdao*, 2019.
- [5] S. Xing et al., "Dual-domain signal fusion adversarial network based Vision Transformer for cross-domain fault diagnosis," *PHM-Hangzhou*, 2023.
- [6] X. Yu et al., "An adaptive domain adaptation method for rolling bearings' fault diagnosis fusing deep convolution and self-attention networks," *IEEE TIM*, 2023.
- [7] Y. Xiao, H. Shao, S. Han, Z. Huo, and J. Wan, "Novel joint transfer network for unsupervised bearing fault diagnosis from simulation domain to experimental domain," *IEEE/ASME TMECH*, 2022.
- [8] R. Guan and Q. Lv, "U-CMG: U-shaped cross-domain control moment gyroscope bearing fault diagnosis method," *CAIT*, 2025.

- [9] H. Wang, C. Li, and Y.-F. Li, "Large-scale visual language model boosted by contrast domain adaptation for intelligent industrial visual monitoring," *IEEE TII*, 2024.